Which of the collected papers can the 3D solver reproduce?

An assessment of the 18 PDFs in `Documents/papers/2026/OT_SB_examples`

Optimal Transport / Schrödinger Bridge Project

5 October 2026

Abstract

Each paper was read against what the solver does today: minimize $\int_0^1 \int_\Omega |m|^2/(2\rho)$ subject to $\partial_t \rho + \nabla \cdot m = \varepsilon \Delta \rho$ with $\varepsilon \geq 0$ constant and isotropic, two hard marginals of equal mass, no-flux walls on $[0,1]^3$, uniform grid with $n_t = n_x = n_y = n_z \leq 96$ in double precision (112 in `mixedc`). Verdicts are **as-is**, **extension: named**, or **not realistic**. Section 1 has the table and the recommended order; Section 2 the unit conversions that matter; Sections 3–5 the per-paper detail. The readers worked from extracted text (no page renderer on this machine), so numbers that exist only inside plots are marked as such.

1 Summary

Paper	ε	Verdict	What we would compare against
Haber, Rehman, Tannenbaum 2010 (SISC)	0	**as-is**	Table 1: ∞-norm map error 1.2e−2, 3.8e−3, 9.0e−4, 2.2e−5, 5.1e−6 at $8^3 \dots 128^3$; 36 projection iterations, mesh-independent
ur Rehman et al. 2009 (MedIA)	0	**as-is**	Fig. 4 $O(h^2)$ map error; 15 min at 128^3 on a 2008 GPU
Lavenant et al. 2024 / Chizat et al. 2022, synthetic tristable SDE	$\sigma^2\Delta t/2$, physical	**as-is**	Lagrangian ground truth from 10^5 particles; hold-one-out W_2 at the midpoint
Chen, Elkin, Tannenbaum 2023 (rOMT), Gaussian spheres	3e−6 (tiny)	**as-is**	Public 50^3 data; closed-form translating diffusing Gaussian; their MSE 0.141 needs the free endpoint
Chen, Georgiou, Pavon 2016, Brownian prior $A = 0$, $B = I$	$s^2/2$	**as-is**	Closed-form $\Sigma(t)$ from Prop. 4 (same family as Bunne 2023)
Aurell et al. 2011, optical trap	0	**as-is**	$W_{\text{diss}} = \tau W_2^2/\Delta t$ with the closed form (24); SB with $\varepsilon > 0$ is a different problem
Reich 2019, drift-free random walk (Ex. 2.5 in 3D)	$\gamma/2$	**as-is**	Exact Schrödinger kernels of a Gaussian-mixture ensemble; constructed, not in the paper
Brenier et al. 2003 / Lévy et al. 2021 (cosmology)	0	**extension: periodic BC or padded sub-box, density floor**	$P(k)$ of reconstructed ICs within 5% for $k < 0.15\,h$/Mpc; 62% of particles within one $6.25\,h^{-1}$Mpc mesh; BAO peak to 0.4%
Chen, Georgiou, Pavon 2018 Part III	$\varepsilon/2$	**extension: potential term $\int V\rho$**	Closed-form Gaussian bridge with harmonic running cost
Chen, Benveniste, Tannenbaum 2024 (urOMT)	3e−6	**extension: unbalanced source + Fisher–Rao**	Public Zenodo data; NMSE / PCTM curves (plots only)
Reich 2019, Lorenz-63 (Ex. 5.3)	$\gamma/2 = 0.05$, physical	**extension: prescribed drift $f(z)$ in the constraint**	Sinkhorn solution of the same single cycle; no numbers in the paper
Holdijk et al. 2023 (transition paths)	$k_BT/(m\gamma)$, derived	**extension: potential term, periodic, basin marginals**	Nothing density-level in the paper; qualitative saddle crossing
Elkin 2018, Koundal 2020, Ratner 2017	$\leq$ 3e−6	**extension: free-endpoint fidelity term**	Private data, non-cubic grids, no numbers to match
Nikakhtar et al. 2022 (BAO, semi-discrete)	0	**not realistic**	Halo-level Laguerre barycentres at 1 Gpc
Hartmann, Schütte 2012	0.5	**not realistic**	Hitting-time control, Dirichlet target set, no second marginal
CGP 2016/2018 own examples	–	**not realistic**	Degenerate anisotropic noise in phase space
gWOT / mean-field Langevin proper	–	**not realistic**	Multi-marginal with soft data fitting in 10-d PCA

Recommended order.

1. **Haber–Rehman–Tannenbaum synthetic Monge–Ampère test** ($\varepsilon = 0$). Only quantitative 3D benchmark with a printed error table and a timing. Needs one post-processing step: extract the map from the dual potential (or integrate $v = m/\rho$) and a floor on zero intensities.
2. **Cosmological reconstruction on AbacusCosmos** ($\varepsilon = 0$). Real public data with initial conditions and particle IDs; the Benamou–Brenier interpolant $\rho(t)$ at $t = D(z{=}49)/D(z{=}0.3)$ *is* their reconstruction, so no particle tracking is needed. Periodic boundaries are a small change to the projection (DCT modes become FFT modes); the padded sub-box works today.
3. **Tristable SDE of Lavenant et al. Sec. 4.1 in 3D** (ε physical). The one problem in the folder where the Brownian reference, two time points and a known σ^2 make our equation exactly their model, with a Lagrangian ground truth to score against.
4. **Brownian-prior Gaussian steering** (CGP Part I with $A = 0$, $B = I$) as a closed-form verification, and **the rOMT Gaussian spheres** (public) as the link to the glymphatic literature.
5. If one extension is to be built, make it the **prescribed drift** $\partial_t\rho + \nabla\cdot(\rho f) + \nabla\cdot m = \varepsilon\Delta\rho$: it unlocks Lorenz-63 data assimilation (the most convincing physical-ε example), a fixed-horizon cousin of Hartmann–Schütte, and the molecular transition-path bridge. The potential-term extension (linear in ρ, changes only the prox) is cheaper and covers CGP Part III and, via Girsanov, gradient drifts.

The honest finding on "ε really needed". Of the applications collected, only the trajectory-inference model and Lorenz-63 assimilation use a diffusion that is both physical and resolved on the grid. In the glymphatic papers $\sigma^2 = 0.002$ in voxel units gives $\varepsilon \approx 3\times10^{-6}$ on our unit box, a regularizer rather than a resolved diffusivity. In Aurell et al. the diffusion cancels: the minimum-dissipation protocol is the deterministic displacement interpolation, and an $\varepsilon > 0$ bridge there answers the variance question they only mention. In cosmology nobody gives $\varepsilon > 0$ a meaning; the one hook is Brenier et al.'s remark that multistream regions have lost their inner velocity structure.

2 Unit conversions

Every paper writes the diffusion differently. With physical domain side L, horizon T, and our equation $\partial_t\rho + \nabla\cdot m = \varepsilon\Delta\rho$ on $[0,1]^3\times[0,1]$:

- Langevin $dX = \sigma\, dB$ (Lavenant, Chizat, Bunne): FP has $\frac{\sigma^2}{2}\Delta\rho$, so $\varepsilon = \sigma^2 T/(2L^2)$.
- Reich $dZ = f\,dt + \gamma^{1/2}dW$: $\varepsilon = \gamma T/(2L^2)$.
- CGP $dx = u\,dt + dw$ with $B = I$: $\varepsilon = T/(2L^2)$; rescaling the Gaussians to width s gives $\varepsilon = s^2/2$.
- rOMT $\partial_t\rho + \nabla\cdot(\rho v) = \sigma\Delta\rho$ (Chen 2023/2024, $\sigma = 0.002$, $T = m\,k_t = 4$, $L = 50$ voxels): $\varepsilon = \sigma T/L^2 = 3.2\times10^{-6}$.
- Entropic OT with $\int \frac{1}{2}|x-y|^2 d\gamma + \epsilon_{\mathrm{OT}} H(\gamma)$: $\epsilon_{\mathrm{OT}} = \sigma^2\Delta t$, so $\varepsilon = \epsilon_{\mathrm{OT}}/(2L^2)$.

3 Reproducible as-is

3.1 Haber, Rehman, Tannenbaum, SIAM J. Sci. Comput. 32 (2010); ur Rehman et al., Med. Image Anal. 13 (2009)

Problem: $\min \frac{1}{2}\int \mu_0|u-x|^2$ s.t. $\det(\nabla u)\,\mu_1(u) = \mu_0$ (eq. 1.1, p. 198), i.e. $\frac{1}{2}W_2^2$, our $\varepsilon = 0$ objective. Intensities are mass; both volumes divided by their sums; a small positive value added where the intensity is zero (value not stated); Dirichlet Laplacian in the inner solves. Synthetic test (both papers): μ_1 = a brain MRI (2009: the built-in MATLAB `mri` volume, $128\times128\times27$), $u = \nabla\varphi$ with $\varphi = \frac{1}{2}|x|^2 + c\prod_i e^{-(x_i-1/2)^2/2\sigma_i^2}$ (c, σ_i not stated), $\mu_0 = \det(\nabla u)\mu_1(u)$; 2010 extends a 2D bump symmetrically in z, which lets our 3D run be checked slice-by-slice against the 2D solver. Targets: 2010 Table 1 (3D ∞-norm map error at $h = 2^{-3}\ldots2^{-7}$: 1.2e−2, 3.8e−3, 9.0e−4, 2.2e−5, 5.1e−6) and Table 2 (projection iterations 67, 46, 34, 35, 36); 2009 Fig. 4 $O(h^2)$ and "15 minutes at 128^3" on a GeForce 8800. Neither reports the transport cost. The real pre-/intra-operative brain pair at 256^3 is private and too large: **not realistic**.

3.2 Lavenant, Zhang, Kim, Schiebinger, Ann. Appl. Probab. 34 (2024); Chizat, Zhang, Heitz, Schiebinger, NeurIPS 2022

Model: $dX = -\nabla\Psi\,dt + \sigma\,dB$ with reflection on a box (Chizat eq. 1) and the reference process is *drift-free* reflected Brownian motion (Lavenant p. 433, 440). Between two consecutive snapshots the estimator is the entropic OT with $\epsilon = \sigma^2\Delta t$ and the Neumann heat kernel (Lavenant eq. 1.6, App. C; Chizat Thm. 3.1 and eq. 6), so the marginal path is exactly $\partial_t\rho + \nabla\cdot m = \frac{\sigma^2}{2}\Delta\rho$ with no-flux walls. The synthetic tristable landscape of Lavenant Sec. 4.1 ($\Psi = 4\|x-x_0\|^2\|x-x_1\|^2\|x-x_2\|^2$ with the three wells in the (x_1, x_2) plane, $X_0 \sim 0.15\,N(0, I)$, $\sigma^2 = 0.25$, horizon 0.4) is 4-d only by padding: dropping x_4 gives a 3D problem unchanged in substance. Minimum experiment: simulate $\geq 10^5$ particles, KDE two snapshots t_i, t_{i+2} onto 96^3, solve the bridge with $\varepsilon = \sigma^2\Delta t/(2L^2)$, compare the $t = \frac{1}{2}$ interpolant with the true $\rho_{t_{i+1}}$ in W_2 (their hold-one-out validation, p. 475). A larger gap raises ε into a regime where the diffusion is visible. Known growth can be folded into one pair by dividing ρ_1 by g_i (hard constraint 3.7). The real data (iPSC reprogramming, 10-d PCA, soft multi-marginal fit) are **not realistic**; a 3-PCA gridding would not be their model.

3.3 Chen, Tran, Elkin, Benveniste, Tannenbaum, J. Sci. Comput. 97 (2023)

Stated model (eqs. 1–2b): $\inf\int\int \rho|v|^2$ s.t. $\partial_t\rho + \nabla\cdot(\rho v) = \sigma\Delta\rho$, equal mass, $\Omega \subseteq \mathbb{R}^3$, explicitly "equivalent to the Schrödinger bridge" (p. 12). This is ours. Solved numerically in free-endpoint form with $\beta\|\rho(1) - \rho_1^{\text{obs}}\|^2$ (eqs. 3–4b) on cell-centred grids, PIC advection + backward-Euler diffusion, Gauss–Newton; Table 1: $m = 10$, $k_t = 0.4$, $\sigma = 0.002$, $\beta = 5000$, grids 50^3 and $56\times106\times51$; run times 3 min / 1.14 h / 3.36 h at $25^3/63^3/88^3$ (12 threads). Public: `https://github.com/xinan-nancy-chen/rOMT_spdup` (Gaussian spheres and two rat cases). Synthetic data (eq. 18): $G_i = \frac{100}{\sqrt{2\pi}}\exp(-[(x-0.8i)^2+(y-0.8i)^2+(z-0.8i)^2]/2)$ on 50^3, blurred with std $(i+1)\sqrt{0.2}$ voxels, so the variance growth ($0.2(2i+3)$ voxel^2 per pair) is inconsistent with $2\sigma T = 0.016$; their optimizer absorbs that through advection and the free endpoint. As-is: the sphere pair with $\varepsilon = 3.2\text{e}{-6}$, or $\varepsilon = 1.6\text{e}{-4}$ to match the actual spreading, which is a closed-form translating diffusing Gaussian. Matching their MSE 0.1412 ± 0.0577 needs the free-endpoint term; the rat crops need per-axis grid sizes.

3.4 Chen, Georgiou, Pavon, IEEE TAC 61 (2016), Brownian-prior case

Their examples use a degenerate noise channel in phase space (**not realistic**). With $A = 0$, $B = I$, $T = 1$ the bridge is $\partial_t\rho + \nabla\cdot(\rho u) = \frac{1}{2}\Delta\rho$ and Prop. 4 (p. 1162) gives $\Sigma(t)^{-1} = P(t)^{-1} + Q(t)^{-1}$ with $Q(t) = Q(0) - tI$, $P(t) = P(0) + tI$, $Q(0) = \Sigma_0^{1/2}\big(\Sigma_0 + \frac{1}{2}I - (\Sigma_0^{1/2}\Sigma_T\Sigma_0^{1/2} + \frac{1}{4}I)^{1/2}\big)^{-1}\Sigma_0^{1/2}$. Scaling the Gaussians to width s gives $\varepsilon = s^2/2$. Minimum experiment: centre $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, $\Sigma_0 = s^2 I \to \Sigma_1 = s^2 I/4$, $s \approx 0.1$, compare $\rho(t)$ at $t = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$; anisotropic $\Sigma_1 = s^2\mathrm{diag}(0.05, 1, 1)$ mirrors their Fig. 4. Same family as the Bunne et al. closed form in the survey.

3.5 Aurell, Mejía-Monasterio, Muratore-Ginanneschi, PRL 106 (2011)

Overdamped Langevin with controllable potential; mean heat eq. (6); FP eq. (8). The optimal control $U^* = -\psi - \beta^{-1}\ln\rho$ cancels the diffusion exactly and the density follows the deterministic system (14)–(15): Burgers for ψ and continuity for ρ, i.e. the W_2 displacement interpolation. Minimal dissipated work $W_{\mathrm{diss}} = \tau W_2^2(\rho_0, \rho_f)/\Delta t$ (eq. 21). Temperature enters only through the entropy difference. Optical-trap example (eqs. 22–24): $U_0 = \frac{1}{2}|x|^2$, $U_f = \frac{1}{2}c|x-h|^2$, Gaussian ansatz valid in any dimension, optimum $\mu_{t_f} = c\Delta t\, h/(c\Delta t + 2\tau)$, $r = K^2/\Delta t^2$, $K = \sqrt{\Delta t(c\Delta t + 2\tau) + \tau^2} - \tau$. As-is with $\varepsilon = 0$: our cost is $\frac{1}{2}W_2^2$, so $W_{\mathrm{diss}} = 2\tau\cdot\mathrm{cost}/\Delta t$. A 3D $\varepsilon > 0$ run is *forced*: it computes the KL / variance-minimizing bridge the paper only mentions (refs. 18–19), not minimum heat.

3.6 Reich, Acta Numerica 28 (2019), random-walk model

The Schrödinger problem is scenario (C): couple the forecast ensemble π_0 at $t = 0$ to the Bayesian analysis $\hat{\pi}_1$ at $t = 1$ with equal weights (Def. 2.4, eq. 2.13; Lemma 2.25 eq. 2.69), reference $dZ = f\,dt + \gamma^{1/2}dW$, path-space cost $\int \|u\|^2/(2\gamma)$ by Girsanov (eq. 2.34). With $\Psi = \mathrm{id}$ (Example 2.5: 11 particles on $[-1, 1]$, $\gamma B = 0.1$, $R = 0.1$, $y = -0.5$) the kernel is Gaussian and $\hat{\pi}_1$ is closed form, so a 3D analogue on $[0, 1]^3$ is as-is with $\varepsilon = \gamma/2$ and exact Schrödinger kernels to compare. It is a constructed example, not an experiment in the paper.

4 Reproducible with a named extension

4.1 Brenier et al., MNRAS 346 (2003); Lévy, Mohayaee, von Hausegger, MNRAS 506 (2021)

Both state the Benamou–Brenier action explicitly: Brenier eq. (51)/(D5), Lévy eq. (9), with growth-factor time $\tau \in [0, 1]$, $\rho_0 \equiv 1$ exactly, quadratic cost, $\varepsilon = 0$. Every simulation box is periodic; Brenier unwraps by $\pm$box and truncates to a sphere, Lévy computes periodic Laguerre diagrams; Brenier's Caffarelli–Li remark (p. 505, 507) justifies a finite patch in a uniform background, which is our Neumann sub-box. Data: AbacusCosmos (public, `https://lgarrison.github.io/AbacusCosmos/`), $1100\,h^{-1}$Mpc boxes, 1440^3 particles, snapshots $z = 0.3 \ldots 1.5$, ICs from `zeldovich-PLT`; Lévy paints 256^3 subsamples on 512^3 with `nbodykit`. Targets: Lévy Fig. 11, reconstructed $z = 49$ power spectrum within 5% for $k \lesssim 0.15\,h/$Mpc; BAO peak $\delta r_s = 0.02 \pm 0.37\%$ over 50 reconstructions (full box only); Brenier Figs. 8–11, 62% of Lagrangian grid points exact and 75% within one mesh at $6.25\,h^{-1}$Mpc, 35% at 3.1, dropping to 45% without the periodicity correction. Needed: periodic BC (DFT modes in the projection; the time solve is unchanged) or a $500\,h^{-1}$Mpc sub-box with Neumann padding of one displacement ($\approx 10\,h^{-1}$Mpc); a density floor for empty voids at 96^3 CIC. Resolution caveat: 96^3 over $1100\,h^{-1}$Mpc gives a Nyquist of $0.27\,h/$Mpc, so only the $k < 0.15$ comparison is meaningful. A natural $\varepsilon > 0$ experiment: compare ε-smearing with their $2\,h^{-1}$Mpc visualization filter and with BAO damping, but no paper gives it a physical reading.

4.2 Chen, Georgiou, Pavon, IEEE TAC 2018, Part III

Adds a running state cost $\int x'Qx$ (eqs. 1a–c), solved in closed form through the Hamiltonian matrix (Thm. 1). For $A = 0$, $B = I$, $Q = qI$ this is our problem plus a potential term $\int\int V\rho$ with $V = q|x|^2$, linear in ρ and therefore only a change to the KE prox. Gives an exact "bridge with harmonic running cost" test in 3D. Their own inertial-particle examples: **not realistic**.

4.3 Chen, Benveniste, Tannenbaum, Sci. Rep. 14 (2024), urOMT

Constraint $\partial_t\rho + \nabla\cdot(\rho v) = \sigma\Delta\rho + \chi\rho r$ with Fisher–Rao cost $\alpha\chi r^2\rho$ (eqs. 3a–c), Neumann via ghost points, free endpoint. Table 1: 50^3 and $56 \times 106 \times 51$, $\sigma = 0.002$, $\alpha = 9000/10000$, $\beta = 5000/50$; run times 3 h 20 min (5 spheres), 8 h 50 min (rat), 3 CPUs. Data public: https://github.com/xinan-nancy-chen/urOMT, Zenodo 10.5281/zenodo.7839019. Needs the unbalanced source term; without it the mass renormalization removes exactly the effect they demonstrate.

4.4 Reich 2019, Lorenz-63 (Example 5.3)

$dZ = f(Z)dt + \gamma^{1/2}dW$, $\gamma = 0.1$, fully observed with $R = 0.1$, $\Delta t = 0.01$ over $t \in [0, 40\,000]$; Fig. 5.1 RMSE vs ensemble size (plot only). A single assimilation cycle is a 3D bridge with physical $\varepsilon = \gamma/2 = 0.05$ in model units on the box containing the attractor, but the reference carries the Lorenz drift, which must enter the constraint as $\nabla\cdot(\rho f)$; the projection is then no longer a pure DCT problem and the drift has to be split off (operator splitting, or absorbed into the LADMM constraint step). Use an observation interval of 0.1–0.5, not the 0.01 step, or ε on the box degenerates to 10^{-6}. The sequential RMSE curves cannot be reproduced by grid bridges.

4.5 Holdijk et al., NeurIPS 2023

Underdamped Langevin in full Cartesian coordinates with the force field as drift, $T = 300$ K, $\gamma = 1\,\mathrm{ps}^{-1}$ (App. C.1); Thm. 3.1 identifies the biased transition-path problem as a Schrödinger bridge with Dirac marginals replaced by a smoothed terminal cost; solved by sampling and a neural policy; Table 1 reports path-level metrics (EPD, THP, ETP) that have no density counterpart. A collective-variable bridge needs: an overdamped reference with $\varepsilon = k_BT/(m\gamma)$ (derived, not stated), the free-energy gradient as drift (by Girsanov equivalent to the Brownian prior plus the potential $|\nabla F|^2/2 - \varepsilon\Delta F$), periodic boundaries for dihedrals, and basin Gaussians as marginals; the 3-CV polyproline free energy is not provided. This would be a new experiment rather than a reproduction.

4.6 Elkin et al. 2018; Koundal et al. 2020; Ratner et al. 2017

Ratner: classical balanced Benamou–Brenier, $\varepsilon = 0$ (eqs. 4–6), equal mass by normalization; qualitative pathway figures only; data private. Elkin (eq. 1, 3): $\partial_t\rho + \nabla\cdot(\rho v) = \sigma^2\Delta\rho$ with a free endpoint $\alpha\|\rho(T) - \rho_T^{\mathrm{obs}}\|^2$; σ reported as 0.002 with an ambiguous square. Koundal (eqs. 1–3): the stated problem is exactly our balanced bridge with $\varepsilon = \sigma^2$, identified with the Schrödinger bridge (p. 12); every reported number (pathline volumes 7124 ± 605 vs 5822 ± 636 voxels, speeds 0.10 ± 0.026 vs 0.09 ± 0.02) comes from the free-endpoint variant plus Lagrangian post-processing on unpublished data with unstated grids; 52 h on a cluster for 14 rats. All three need the free-endpoint term to match anything and none has public data.

5 Not realistic

- **Nikakhtar, Sheth, Lévy, Mohayaee, PRL 129 (2022).** Weighted semi-discrete OT to halo centres plus dust on a $1\,h^{-1}$Gpc periodic box, 20 realizations, sub-percent BAO statistics; a 96^3 Eulerian grid cannot resolve protohalos. A crude mass-weighted-halo analogue is possible but compares to nothing in the paper.
- **Hartmann, Schütte, J. Stat. Mech. (2012).** Rare-event control to a target set with a Dirichlet boundary and stopping time (eqs. 3.2–3.7); 1D double well, $\varepsilon = 0.5$. Not a two-marginal bridge. A fixed-horizon cousin between the two wells would need the drift extension and has no counterpart in the paper.
- **CGP 2016/2018 own examples.** Inertial particles with noise only in velocity ($B = [0; 1]$) in unbounded phase space: degenerate anisotropic diffusion breaks the isotropic projection.
- **gWOT and mean-field Langevin proper.** Fifty soft marginals, growth, 10-d PCA, particle methods.